page 5 :

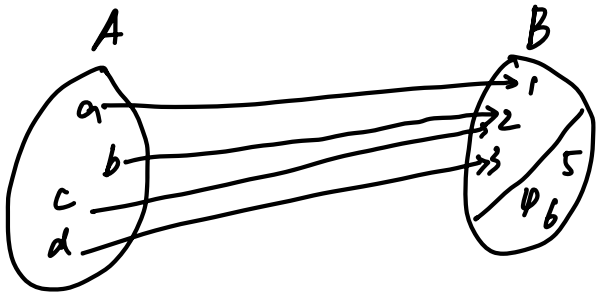
two sets : A , B

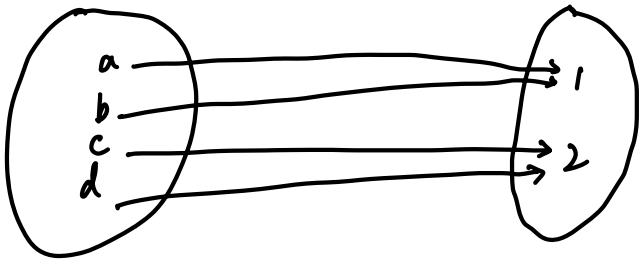
$F : A \to B$

$a \mapsto F(a)$

domain	codomain
A	B
argument	image (element)
a	$F(a)$

$$F(A) = \{ F(a) \mid a \in A \} \subseteq B$$





(onto)

surjective

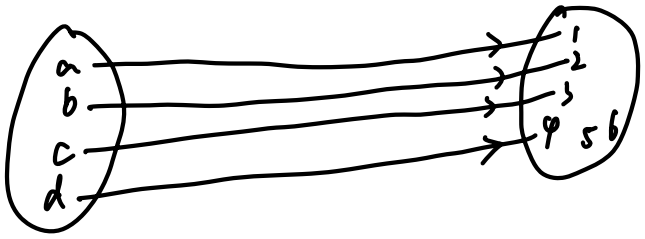
滿射

$F(A) = B$

every element of the codomain is the image of at least one element of the domain

(one-to-one) mapping

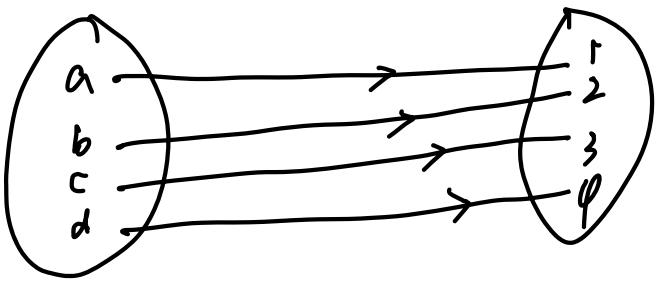
injective



every element of the codomain is the image of at most one element of the domain

one-to-one correspondence

bijective



surjective and injective

one-to-one mapping : if $F(x) = F(y)$

then $x = y$

(codomain)

onto mapping : every element in the range of the mapping has at least one corresponding element in the domain.

(preimage)

From geometry, topology, and physics

P 66.

Definition 2.1. If a map satisfies a certain condition it bears a special name.

(a) A map $f: X \to Y$ is called injective (or one to one) if $x \neq x'$ implies $f(x) = f(x')$.

(b) A map $f: X \to Y$ is called surjective (or onto) if for each $y \in Y$ there exists at least one element $x \in X$ such that $f(x) = y$.

(c) A map $f: X \to Y$ is called bijective if it is both injective and surjective.